

Finite-Dimensional Combinatorial and Arithmetic Structures of Parity Vectors for the Accelerated Collatz Map

Kazunobu Hikawa*

Abstract

We investigate finite-dimensional combinatorial and arithmetic structures of parity vectors associated with the accelerated Collatz map.

In addition to the conventional classification by the sequence length k , we introduce a weight-based framework using the Hamming weight d . This framework induces a natural organization of parity-vector spaces and admits a geometric interpretation on the (k, d) lattice plane.

Our principal result is the rigorous establishment of the structural identity $X(d+1) = W(d)$, where $W(d) = |U(d)|$ denotes the number of unconverged parity vectors of weight d , and $X(d+1) = |J(d+1)|$ denotes the number of just-converged parity vectors of weight $d+1$. The identity is proved by constructing an explicit bijection $\Phi : U(d) \rightarrow J(d+1)$ and establishing both its injectivity and surjectivity.

We further investigate arithmetic properties of the correction term appearing in the fundamental Collatz identity. In particular, for parity vectors of equal length and equal Hamming weight, we derive an exact formula for the 2-adic valuation of the difference between their correction terms. The valuation is shown to be completely determined by the earliest discrepant index of the corresponding parity vectors. This result reveals an arithmetic rigidity within each fixed (k, d) layer of the parity-vector space.

Large-scale computations up to $d = 10,000$ are also presented. The normalized relative density within the minimum admissible binary space is observed to decrease substantially as the Hamming weight increases.

Taken together, these results provide complementary combinatorial and arithmetic descriptions of parity-vector spaces within the weight-based framework. The geometric interpretation developed in this paper visualizes the organization of these structures on the (k, d) lattice plane. In particular, the explicit bijection underlying the structural identity and the 2-adic rigidity theorem within fixed (k, d) layers establish two distinct structural properties of finite parity-vector spaces.

Keywords: Collatz conjecture, parity vector, weight-based classification, structural identity, 2-adic valuation, finite-dimensional structure

*Professor Emeritus, Kanazawa Gakuin University. Email: hikawa@kanazawa-gu.ac.jp

1 Introduction

The Collatz conjecture is one of the most famous unsolved problems in number theory. For the accelerated Collatz map

$$T(n) = \begin{cases} (3n+1)/2, & n \equiv 1 \pmod{2}, \\ n/2, & n \equiv 0 \pmod{2}, \end{cases}$$

the conjecture asserts that every positive integer eventually reaches 1 under iteration.

The primary objective of the present paper is to investigate finite-dimensional combinatorial and arithmetic structures of parity-vector spaces associated with the accelerated Collatz map. Our principal mathematical contributions are the establishment of a structural identity through an explicit bijection and the derivation of a 2-adic rigidity theorem within fixed (k, d) layers. A geometric interpretation of these structures is also developed on the (k, d) lattice plane.

Throughout this paper, the term “finite-dimensional” refers to parity-vector structures defined at finite sequence lengths and finite Hamming weights, in contrast to infinite parity sequences or the global dynamics of Collatz orbits.

Parity vectors constitute one of the classical tools for studying the Collatz map. Terras [1] investigated the statistical distribution of parity vectors, and Lagarias [2] established the injectivity theorem showing that a parity vector of length k uniquely determines the residue class of the initial value modulo 2^k . Simons and de Weger [3] derived theoretical and computational bounds for nontrivial cycles of the $3x+1$ problem, providing further structural restrictions on possible periodic behaviors.

General surveys and historical developments are presented in Lagarias [4, 6] and Martiny [5]. Computational verifications have been extended up to 2^{71} by Bařina [8], in agreement with the extensive computations of Roosendaal [9]. More recently, Tao [7] proved that almost all orbits of the Collatz map attain almost bounded values, providing a major advance from the viewpoint of analytic and probabilistic methods.

Unlike these analytical and dynamical approaches, the present work focuses on finite-dimensional combinatorial and arithmetic structures induced by parity vectors. This work represents the third stage of a series of investigations on parity-vector structures associated with the Collatz map.

In Hikawa–Nakanishi [10], a classification based on the Hamming weight d , that is, the number of ones in a parity vector, was introduced in addition to the conventional classification by the sequence length k . Within this framework, the counting relation

$$X(d+1) = W(d)$$

was discovered through large-scale computations and reported as an empirical structural identity. Recursive algorithms for generating parity vectors satisfying the necessary conditions for unconverged growth were also developed.

The underlying prefix inequality expresses the necessary requirement that the cumulative multiplicative growth produced by odd steps dominate the cumulative halving effect at every prefix. More precisely, a parity vector is classified as unconverged when every prefix satisfies

$$j < \lambda d(V|_j), \quad \lambda = \log_2 3,$$

or equivalently,

$$d(V|_j) > \frac{j}{\lambda}.$$

Thus, the term “unconverged” refers to the continued satisfaction of this prefix condition throughout the finite parity vector. Consequently, any hypothetical infinite non-convergent trajectory must satisfy the corresponding condition at every finite prefix.

Subsequently, Nakanishi [11] investigated the same weight-based framework from a structural viewpoint. By means of mathematical induction, partition properties of parity vectors and the absence of critical substructures were established. These results clarified several combinatorial aspects of parity-vector spaces. However, neither a set-theoretic formulation nor an explicit construction of the correspondence underlying

$$X(d+1) = W(d)$$

was provided. Consequently, the combinatorial mechanism underlying the empirical structural identity remained unresolved.

The primary purpose of the present paper is to complete this line of investigation by establishing a rigorous set-theoretic formulation of the weight-based classification. We introduce the sets $U(d)$ and $J(d)$, construct an explicit mapping

$$\Phi : U(d) \rightarrow J(d+1),$$

and prove that this mapping is bijective. Consequently, the previously observed relation

$$X(d+1) = W(d)$$

is established as a finite-dimensional combinatorial theorem.

We further investigate arithmetic properties of the correction term appearing in the fundamental Collatz identity. For parity vectors of equal length and equal Hamming weight, we derive an exact formula for the 2-adic valuation of the difference between their correction terms. The valuation is completely determined by the earliest position at which the corresponding parity vectors differ. This result reveals an arithmetic rigidity within each fixed (k, d) layer of the parity-vector space.

The main contributions of the present paper are summarized as follows.

1. We provide a rigorous set-theoretic formulation of the weight-based classification by introducing the sets $U(d)$ and $J(d)$ and constructing an explicit bijection

$$\Phi : U(d) \rightarrow J(d+1).$$

2. We establish the structural identity

$$X(d+1) = W(d)$$

as a finite-dimensional combinatorial theorem.

3. For parity vectors of equal length and equal Hamming weight, we derive an exact formula for the 2-adic valuation of differences between their correction terms. The valuation depends solely on the earliest position at which the two parity vectors differ, thereby revealing an arithmetic rigidity within each fixed (k, d) layer.
4. We present large-scale numerical computations up to

$$d = 10,000,$$

and examine the behavior of the normalized relative density within the proposed weight-based framework.

5. We develop a geometric interpretation of the finite-dimensional structures on the (k, d) lattice plane, providing a visual description of the combinatorial classification and the structural correspondence between neighboring weight layers.

The results established in this paper concern finite parity-vector structures and are derived independently of the truth or falsity of the Collatz conjecture. The explicit bijection describes a combinatorial correspondence between neighboring weight layers, while the 2-adic rigidity theorem reveals an arithmetic structure within fixed (k, d) layers. The geometric interpretation on the (k, d) lattice plane provides a common framework for visualizing these structures.

Although every hypothetical infinite non-convergent trajectory must satisfy the un-converged prefix condition at every finite stage, the finite-dimensional results established here do not by themselves determine the existence or nonexistence of such an infinite trajectory. Their implications for the global dynamics of the accelerated Collatz map therefore remain an open problem.

The remainder of the paper is organized as follows. Section 2 introduces the basic formulation and notation. Section 3 constructs the explicit bijection and proves the structural identity. Section 4 presents a geometric interpretation on the (k, d) lattice plane. Section 5 studies arithmetic properties of correction terms and establishes the 2-adic rigidity theorem within fixed (k, d) layers. Section 6 presents numerical results. Section 7 discusses open problems and future directions. Section 8 concludes the paper.

2 Foundational Formulation

2.1 Accelerated Collatz Map and Parity Vectors

Throughout this paper, we consider the accelerated Collatz map

$$T : \mathbb{N} \rightarrow \mathbb{N}$$

defined by

$$T(n) = \begin{cases} \frac{3n+1}{2}, & n \equiv 1 \pmod{2}, \\ \frac{n}{2}, & n \equiv 0 \pmod{2}. \end{cases} \quad (2.1)$$

In the odd case, the operations $3n+1$ and the subsequent division by 2 are regarded as a single iteration. Thus, the map defined by (2.1) is the standard accelerated Collatz map.

For an initial value $n_0 \in \mathbb{N}$, let

$$n_j = T^j(n_0), \quad 0 \leq j \leq k,$$

denote the trajectory generated by T .

For each step $j = 1, \dots, k$, define

$$v_j = \begin{cases} 1, & n_{j-1} \equiv 1 \pmod{2}, \\ 0, & n_{j-1} \equiv 0 \pmod{2}. \end{cases} \quad (2.2)$$

The binary sequence

$$v^{(k)} = (v_1, \dots, v_k) \in \{0, 1\}^k$$

is called the *parity vector* (PV) of length k .

The Hamming weight of a parity vector $v^{(k)}$ is denoted by $d(v^{(k)})$, where

$$d(v^{(k)}) = \sum_{j=1}^k v_j. \quad (2.3)$$

When no ambiguity arises, we simply write $d(k)$ instead of $d(v^{(k)})$.

Since the present study focuses on structures classified by the number of odd steps, the quantity $d(k)$ plays a central role.

For later convenience, we introduce the constant

$$\lambda := \log_2 3 = \frac{\log 3}{\log 2} \approx 1.58496. \quad (2.4)$$

The critical boundary on the (k, d) lattice plane is therefore expressed as

$$k = \lambda d. \quad (2.5)$$

2.2 Fundamental Identity and the Correction Term

Repeated application of the accelerated Collatz map yields the classical identity

$$n_k = \frac{3^{d(k)}}{2^k} n_0 + q_k(v^{(k)}), \quad (2.6)$$

where

$$q_k(v^{(k)}) = \sum_{j=1}^k \frac{v_j 3^{d(k)-d(j)}}{2^{k-j+1}}, \quad (2.7)$$

and

$$d(j) = \sum_{i=1}^j v_i. \quad (2.8)$$

The quantity

$$q_k(v^{(k)})$$

will be referred to as the *correction term* associated with the parity vector. Its arithmetic properties constitute one of the principal subjects of the present paper.

Multiplying (2.6) by 2^k gives

$$2^k n_k = 3^{d(k)} n_0 + 2^k q_k(v^{(k)}), \quad (2.9)$$

which shows that

$$2^k q_k(v^{(k)}) \in \mathbb{Z}. \quad (2.10)$$

Consequently, a necessary congruence condition for realizability is

$$3^{d(k)}n_0 \equiv -2^k q_k(v^{(k)}) \pmod{2^k}. \quad (2.11)$$

Since

$$\gcd(3, 2) = 1,$$

there exists a unique residue class

$$n_0 \pmod{2^k}$$

corresponding to every parity vector $v^{(k)} \in \{0, 1\}^k$.

This uniqueness property is essentially equivalent to the injectivity theorem established by Lagarias [2], which states that every parity vector of length k uniquely determines the residue class of the initial value modulo 2^k .

The finite-dimensional combinatorial and arithmetic structures developed in the subsequent sections are built upon the fundamental definitions introduced in this section.

3 The Structural Identity $X(d+1) = W(d)$

This section establishes the principal combinatorial result of the present paper. By introducing the finite-dimensional sets $U(d)$ and $J(d)$ and constructing an explicit bijection

$$\Phi : U(d) \rightarrow J(d+1),$$

we prove the structural identity

$$X(d+1) = W(d),$$

thereby providing a rigorous set-theoretic foundation for the proposed weight-based framework.

3.1 Definition of the Sets

Let

$$P^{(1)} = \bigcup_{k \geq 1} \{V \in \{0, 1\}^k \mid v_1 = 1\}$$

denote the set of all parity vectors beginning with 1.

For $V \in P^{(1)}$, let

$$|V| = k$$

denote its length,

$$d(V) = \sum_i v_i$$

its Hamming weight, and

$$V|_j$$

its prefix of length j .

Since

$$\frac{3^{d(V|_j)}}{2^j} > 1$$

is a necessary condition for the cumulative growth produced by odd steps to dominate the cumulative halving effect up to the j -th step, one obtains the prefix inequality

$$j < \lambda d(V|_j), \quad \lambda = \log_2 3. \quad (3.1)$$

This condition represents the weakest prefixwise requirement for unconverged growth and motivates the following definitions.

We define

$$U = \{V \in P^{(1)} \mid j < \lambda d(V|_j) \text{ for all } 1 \leq j \leq |V|\}, \quad (3.2)$$

and, for each weight level d ,

$$U(d) = \{V \in U \mid d(V) = d\}, \quad (3.3)$$

together with

$$J(d) = \{W \in P^{(1)} \mid d(W) = d, |W| \geq \lambda d, W|_{|W|-1} \in U\}. \quad (3.4)$$

Since $\lambda = \log_2 3$ is irrational, the critical boundary never passes through a lattice point. Consequently, every parity vector in $J(d)$ strictly crosses the critical boundary at its terminal step.

Their cardinalities are denoted by

$$W(d) = |U(d)|, \quad X(d) = |J(d)|. \quad (3.5)$$

Elements of $U(d)$ remain strictly to the left of the critical line

$$k = \lambda d,$$

whereas elements of $J(d)$ correspond to parity vectors whose terminal step produces the first crossing of this boundary.

Since $\lambda = \log_2 3$ is irrational, the quantity λd is never an integer. Therefore,

$$|W| \geq \lambda d$$

is equivalent to

$$|W| \geq \lceil \lambda d \rceil.$$

Thus, the condition

$$W|_{|W|-1} \in U$$

guarantees that the parity vector remains inside the unconverged region up to the penultimate step and crosses the critical boundary for the first time at its terminal step.

The sets $U(d)$ and $J(d)$ are finite-dimensional combinatorial objects defined solely through the prefix inequality (3.1), independently of the truth or falsity of the Collatz conjecture.

For later use, we define

$$k_{\min}(d) := \lceil \lambda d \rceil,$$

which represents the smallest admissible length associated with weight d .

3.2 Construction of the Bijection

The following lemma provides the key step for constructing an explicit correspondence between the sets $U(d)$ and $J(d+1)$.

Lemma 3.1. *Let $V \in U(d)$.*

1. *The sequence*

$$(V, 1)$$

belongs to $U(d+1)$.

2. *There exists a unique integer*

$$m \geq 1$$

such that

$$(V, 1, 0^m)$$

belongs to $J(d+1)$.

Proof. (1) Every proper prefix of $(V, 1)$ having length at most $|V|$ coincides with a prefix of V . Since $V \in U(d)$, the defining inequality

$$j < \lambda d_j$$

holds for all such prefixes.

It therefore remains only to verify the defining inequality for the full vector $(V, 1)$. Since

$$|V| < \lambda d,$$

we obtain

$$|(V, 1)| = |V| + 1 < \lambda d + 1 < \lambda(d+1),$$

because

$$\lambda - 1 > 0.$$

Hence the defining inequality also holds for the full vector $(V, 1)$, and therefore

$$(V, 1) \in U(d+1).$$

(2)

Let

$$L_m = |V| + 1 + m.$$

Then

$$(V, 1, 0^m) \in J(d+1)$$

is equivalent to

$$\lambda(d+1) \leq L_m < \lambda(d+1) + 1.$$

This can be rewritten as

$$A \leq m < A + 1,$$

where

$$A = \lambda(d+1) - |V| - 1.$$

Since λ is irrational, the number A is not an integer. Therefore, there exists a unique integer satisfying the above inequality, namely

$$m = \lfloor A \rfloor + 1.$$

Since A is not an integer, we have

$$\lfloor A \rfloor + 1 = \lfloor A + 1 \rfloor.$$

Hence

$$m = \lfloor \lambda(d+1) - |V| \rfloor.$$

Furthermore, since

$$|V| < \lambda d,$$

we obtain

$$\lambda(d+1) - |V| > \lambda(d+1) - \lambda d = \lambda > 1.$$

Therefore,

$$m = \lfloor \lambda(d+1) - |V| \rfloor \geq 1.$$

Thus, the integer m is uniquely determined and satisfies

$$m \geq 1.$$

□

Lemma 3.1 enables us to define the mapping

$$\Phi : U(d) \rightarrow J(d+1)$$

by

$$\boxed{\Phi(V) = (V, 1, 0, \dots, 0), \quad m = \lfloor \lambda(d+1) - |V| \rfloor.} \quad (3.6)$$

The mapping Φ appends a terminal symbol 1 followed by the unique number of zeros required for the parity vector to cross the critical boundary for the first time.

3.3 Main Theorem

We are now ready to establish the principal combinatorial result of the present paper.

Theorem 3.2 (Structural Identity and Explicit Bijection). *The mapping*

$$\Phi : U(d) \rightarrow J(d+1)$$

is bijective.

Consequently,

$$X(d+1) = W(d). \tag{3.7}$$

Proof. Injectivity is immediate from the construction of Φ , since removing the appended terminal block of zeros together with the immediately preceding terminal symbol 1 uniquely reconstructs the original parity vector.

To prove surjectivity, let

$$W \in J(d+1)$$

be arbitrary.

Suppose that the terminal symbol of W were equal to 1.

Then

$$d(W|_{|W|-1}) = d,$$

and, since

$$W|_{|W|-1} \in U,$$

we have

$$|W| - 1 < \lambda d.$$

On the other hand,

$$W \in J(d+1)$$

implies

$$|W| \geq \lambda(d+1).$$

Therefore,

$$|W| - 1 \geq \lambda(d+1) - 1 = \lambda d + (\lambda - 1) > \lambda d,$$

because

$$\lambda > 1.$$

This contradicts

$$|W| - 1 < \lambda d.$$

Hence the terminal symbol of W cannot be 1.

Therefore every element of $J(d+1)$ necessarily ends with a terminal zero.

We now use this fact to prove the surjectivity of Φ .

Since W ends with 0, it possesses a unique maximal terminal block of zeros.

The symbol immediately preceding this block cannot be 0, for otherwise the terminal zero block could be extended by one position to the left, contradicting its maximality.

Hence the symbol immediately preceding the maximal terminal zero block must be 1.

Removing the maximal terminal zero block together with this preceding terminal 1 yields a parity vector

$$V.$$

Every prefix of V is also a proper prefix of W .

Since

$$W|_{|W|-1} \in U,$$

every such prefix satisfies the defining inequality of U .

Hence,

$$V \in U(d).$$

By Lemma 3.1(2), the number of removed terminal zeros is uniquely determined.

Therefore,

$$\Phi(V) = W,$$

which proves that Φ is surjective.

Consequently, Φ is bijective.

Therefore,

$$|U(d)| = |J(d+1)|.$$

Hence, by the definitions of the counting functions,

$$X(d+1) = W(d),$$

which establishes the structural identity.

□

4 Geometric Interpretation

This section provides a geometric interpretation of the finite-dimensional combinatorial structures established in Section 3. The geometric viewpoint serves to visualize the explicit bijection and the structural identity

$$X(d+1) = W(d),$$

rather than to introduce additional assumptions concerning the global dynamics of the accelerated Collatz map.

4.1 Tripartition of the Parity Space

For a fixed Hamming weight d , consider the layer

$$P^{(1)}(d) = \{V \in P^{(1)} \mid d(V) = d\}. \quad (4.1)$$

This space admits the decomposition

$$P^{(1)}(d) = U(d) \cup J(d) \cup A(d), \quad (4.2)$$

where

$$A(d) = P^{(1)}(d) \setminus (U(d) \cup J(d)). \quad (4.3)$$

Since these three sets are mutually disjoint, we obtain

$$P^{(1)}(d) = U(d) \sqcup J(d) \sqcup A(d). \quad (4.4)$$

Geometrically, these subsets admit a natural interpretation on the (k, d) lattice plane. Each subset corresponds to a distinct stage relative to the critical boundary

$$k = \lambda d.$$

- The set $U(d)$ consists of parity vectors whose every prefix satisfies the prefix inequality (3.1). Consequently, every corresponding lattice path remains entirely inside the unconverged region.
- The set $J(d)$ consists of parity vectors whose terminal step produces the first crossing of the critical boundary. Since $\lambda = \log_2 3$ is irrational, the boundary never passes through a lattice point, and therefore every element of $J(d)$ strictly crosses the boundary at its terminal step.
- The set $A(d)$ consists of parity vectors having a proper prefix that has already crossed the critical boundary. These vectors therefore belong to the already-converged region.

Thus, the parity-vector space is partitioned into three finite-dimensional combinatorial regions according to their relative positions with respect to the critical boundary. This tripartition provides the geometric framework used throughout the remainder of this section.

4.2 Geometric Meaning of the Bijection

The explicit mapping

$$\Phi : U(d) \rightarrow J(d+1)$$

constructed in Theorem 3.2 admits a natural geometric interpretation on the (k, d) lattice plane.

Given an unconverged parity vector

$$V \in U(d),$$

one first appends a terminal symbol 1, thereby increasing the Hamming weight from d to $d + 1$ while preserving the prefix inequality established in Lemma 3.1.

A uniquely determined number of terminal zeros is then appended until the lattice path crosses the critical boundary for the first time. Thus,

$$\Phi(V) = (V, 1, 0^m)$$

represents the unique just-converged parity vector generated from the unconverged vector V .

Consequently, the structural identity

$$X(d + 1) = W(d)$$

may be viewed geometrically as a one-to-one correspondence between the unconverged layer of weight d and the just-converged layer of weight $d + 1$. Rather than merely relating their cardinalities, the bijection establishes an explicit structural correspondence between these two finite-dimensional parity-vector spaces.

The geometric tripartition introduced above provides a natural setting in which this correspondence can be visualized.

4.3 The (k, d) Lattice Plane

The finite-dimensional parity-vector space admits a natural visualization on the (k, d) lattice plane, where the horizontal axis represents the sequence length k and the vertical axis represents the Hamming weight d .

The critical boundary

$$k = \lambda d$$

separates the unconverged region from the already-converged region.

Parity vectors belonging to $U(d)$ remain entirely inside the unconverged region throughout all prefixes, whereas vectors belonging to $J(d)$ cross the critical boundary exactly once, namely at their terminal step.

Accordingly, the structural identity

$$X(d + 1) = W(d)$$

admits a geometric interpretation as a discrete correspondence between neighboring weight layers induced by the first crossing of the critical boundary.

Figure 1 summarizes this geometric interpretation of the parity space. The figure illustrates the decomposition

$$P^{(1)}(d) = U(d) \sqcup J(d) \sqcup A(d),$$

together with the explicit bijection

$$\Phi : U(d) \rightarrow J(d + 1),$$

which links the unconverged and just-converged layers.

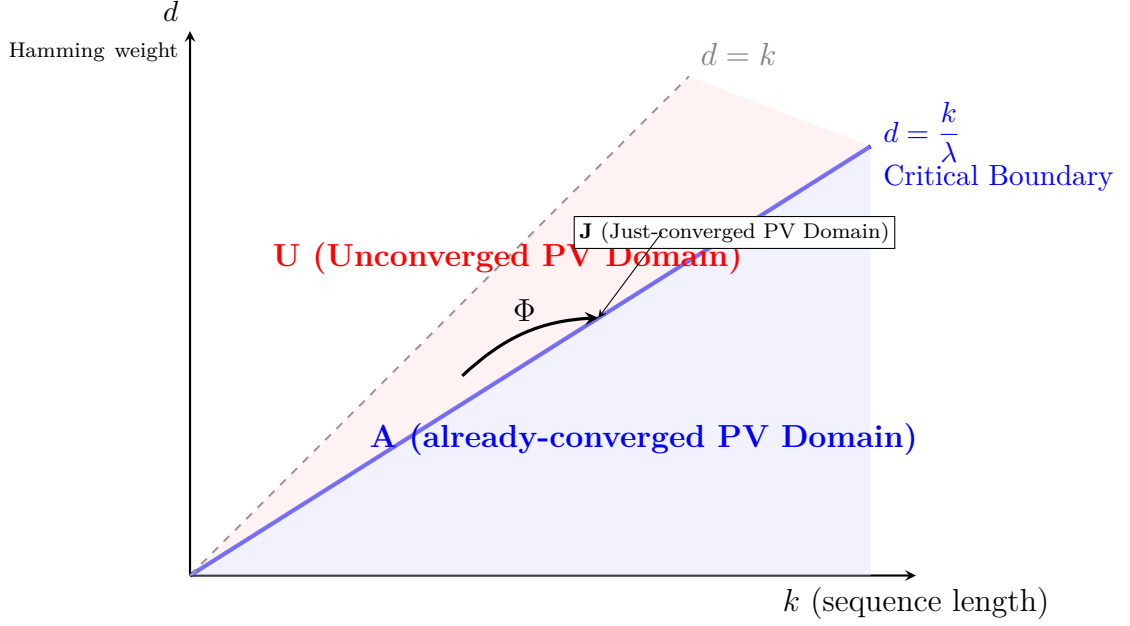


Figure 1: Schematic illustration of the parity space on the (k, d) lattice plane. The critical boundary is shown in the equivalent form $d = k/\lambda$, corresponding to the expression $k = \lambda d$ used throughout the text. This boundary separates the unconverged parity-vector region **U** (upper-left) from the already-converged region **A** (lower-right). The just-converged region **J** consists of parity vectors whose terminal step produces the first crossing of the critical boundary. The arrow labeled Φ illustrates the explicit bijection established in Theorem 3.2.

4.4 Typical Trajectories of Parity Vectors

The geometric classification developed in the preceding sections can be illustrated by representative lattice paths associated with parity vectors.

Recall that every parity vector

$$v = (v_1, v_2, \dots)$$

determines a lattice path on the (k, d) lattice plane.

At each step,

$$v_i = 0$$

corresponds to the horizontal move

$$(k, d) \longrightarrow (k + 1, d),$$

whereas

$$v_i = 1$$

corresponds to the diagonal move

$$(k, d) \longrightarrow (k + 1, d + 1).$$

Consequently, every parity vector may be regarded as a discrete lattice path whose behavior is determined by its position relative to the critical boundary

$$k = \lambda d, \quad \lambda = \log_2 3.$$

Figure 2 illustrates three representative classes of parity vectors.

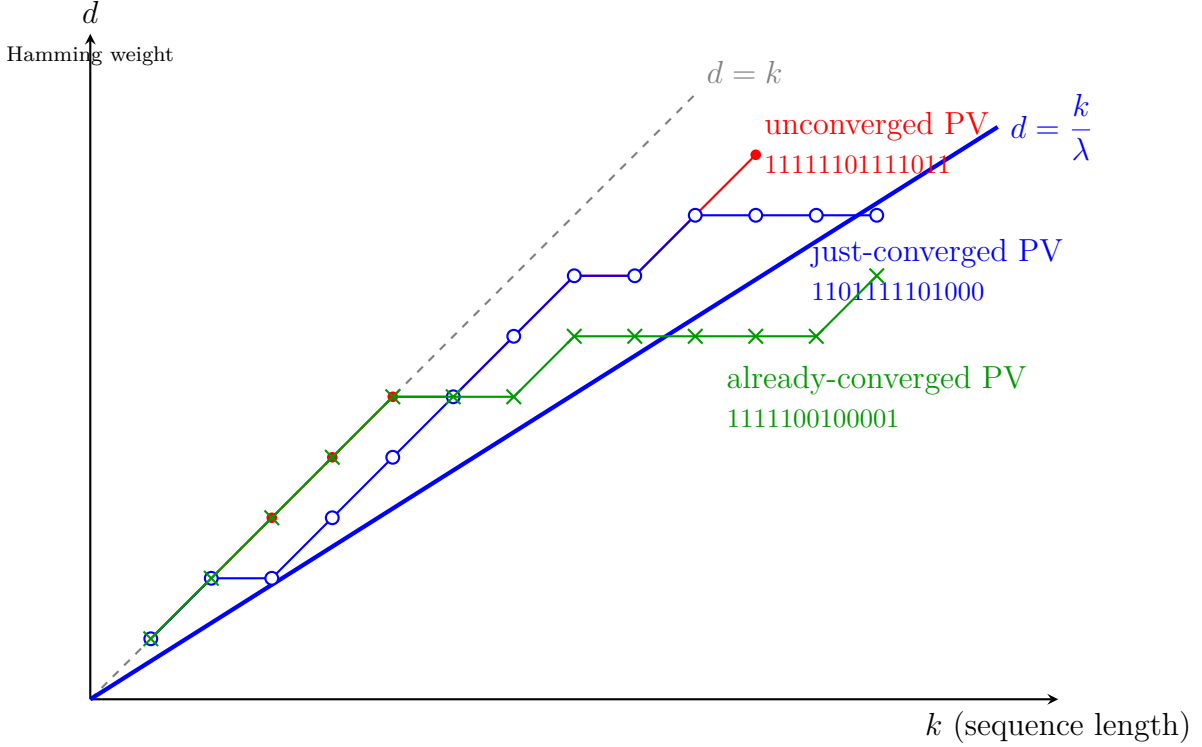


Figure 2: Typical lattice paths associated with parity vectors. An unconverged parity vector remains entirely inside the unconverged region. A just-converged parity vector reaches the critical boundary for the first time at its terminal point. An already-converged parity vector initially stays above the boundary, but eventually passes through the boundary and enters the already-converged region. The representative parity vectors shown in the figure were verified computationally and are intended only as illustrative examples.

- An unconverged parity vector remains entirely inside the unconverged region throughout its trajectory and never crosses the critical boundary.
- A just-converged parity vector crosses the critical boundary for the first time at its terminal step.
- An already-converged parity vector initially remains inside the unconverged region but crosses the critical boundary at an intermediate step and subsequently continues inside the already-converged region.

Consequently, the tripartition

$$P^{(1)} = U \sqcup J \sqcup A$$

admits a simple geometric interpretation in terms of lattice paths on the (k, d) plane.

The representative parity vectors shown in Figure 2 provide concrete examples of these three finite-dimensional combinatorial classes and illustrate how the structural classification established in Section 3 is reflected geometrically.

5 Arithmetic Properties of the Correction Term

The preceding sections established finite-dimensional combinatorial and geometric structures associated with parity vectors. In this section, we investigate an independent arithmetic aspect of the same finite-dimensional parity-vector spaces through the correction term appearing in the fundamental identity (2.6).

Our main objective is to show that differences between correction terms possess a rigid 2-adic structure within fixed (k, d) layers. In particular, we prove that, for parity vectors of the same length and the same Hamming weight, the 2-adic valuation of the difference between their correction terms is determined solely by the earliest position at which the corresponding parity vectors differ. This arithmetic rigidity complements the combinatorial structure established in Section 3.

5.1 Positivity of the Correction Term

Multiplying (2.7) by 2^k , we define the scaled correction term

$$Q_k := 2^k q_k(v^{(k)}). \quad (5.1)$$

Using (2.7), we obtain

$$Q_k = \sum_{j=1}^k v_j 3^{d(k)-d(j)} 2^{j-1}. \quad (5.2)$$

Since every parity vector considered throughout this paper belongs to $P^{(1)}$, its first component satisfies

$$v_1 = 1.$$

Consequently, every term appearing in the defining sum of $q_k(v^{(k)})$ is nonnegative, while the first term is strictly positive.

Therefore, the correction term is strictly positive for every parity vector.

Remark 5.1. For every parity vector,

$$q_k(v^{(k)}) > 0. \quad (5.3)$$

Although elementary, this positivity property will be useful in the subsequent arithmetic analysis of correction terms.

5.2 2-adic Rigidity

Let

$$H_k = \{(v_1, \dots, v_k) \in \{0, 1\}^k \mid v_1 = 1\}.$$

For a parity vector

$$v^{(k)} = (v_1, \dots, v_k) \in H_k,$$

we define the partial Hamming weight by

$$d_v(j) = \sum_{i=1}^j v_i, \quad 1 \leq j \leq k,$$

and write

$$d_v = d_v(k)$$

for its total Hamming weight.

Similarly, for

$$w^{(k)} = (w_1, \dots, w_k) \in H_k,$$

we write

$$d_w(j) = \sum_{i=1}^j w_i, \quad d_w = d_w(k).$$

With this notation, the scaled correction term defined in (5.1)–(5.2) can be written as

$$Q_k(v^{(k)}) = 2^k q_k(v^{(k)}) = \sum_{j=1}^k v_j 3^{d_v - d_v(j)} 2^{j-1}. \quad (5.4)$$

For two distinct parity vectors

$$v^{(k)}, w^{(k)} \in H_k,$$

let

$$j_{\min} = \min \{j \in \{1, \dots, k\} \mid v_j \neq w_j\}$$

denote the earliest index at which the two vectors differ.

Since every element of H_k begins with 1, we necessarily have

$$j_{\min} \geq 2.$$

Throughout this section, $\nu_2(n)$ denotes the 2-adic valuation of a nonzero integer n , namely the largest nonnegative integer a such that

$$2^a \parallel n.$$

For completeness, we adopt the convention

$$\nu_2(0) = +\infty.$$

The following theorem gives an exact evaluation of the 2-adic valuation of the difference between correction terms associated with parity vectors belonging to the same (k, d) layer.

Theorem 5.2 (2-adic Rigidity). *Let*

$$v^{(k)}, w^{(k)} \in H_k$$

be distinct parity vectors of equal Hamming weight,

$$d_v = d_w = d.$$

Then

$$\nu_2 \left(2^k \left(q_k(v^{(k)}) - q_k(w^{(k)}) \right) \right) = j_{\min} - 1. \quad (5.5)$$

Consequently,

$$\left| q_k(v^{(k)}) - q_k(w^{(k)}) \right| \geq 2^{-(k-j_{\min}+1)}. \quad (5.6)$$

Proof. By the equal-weight assumption

$$d_v = d_w = d,$$

the difference of the scaled correction terms can be written term by term as

$$Q_k(v^{(k)}) - Q_k(w^{(k)}) = \sum_{j=1}^k \delta_j 2^{j-1}, \quad (5.7)$$

where

$$\delta_j = v_j 3^{d-d_v(j)} - w_j 3^{d-d_w(j)}. \quad (5.8)$$

We analyze the terms according to their position relative to $j_{\min}$.

Step 1: Indices $j < j_{\min}$.

For every

$$j < j_{\min},$$

the prefixes of $v^{(k)}$ and $w^{(k)}$ of length j coincide. Therefore,

$$v_j = w_j$$

and

$$d_v(j) = d_w(j).$$

Hence,

$$\delta_j = 0.$$

Thus, all terms preceding the first discrepant index vanish.

Step 2: The index $j = j_{\min}$.

At the first discrepant index, exactly one of

$$v_{j_{\min}}, \quad w_{j_{\min}}$$

is equal to 1, while the other is equal to 0.

If

$$v_{j_{\min}} = 1, \quad w_{j_{\min}} = 0,$$

then

$$\delta_{j_{\min}} = 3^{d-d_v(j_{\min})},$$

which is odd.

Similarly, if

$$v_{j_{\min}} = 0, \quad w_{j_{\min}} = 1,$$

then

$$\delta_{j_{\min}} = -3^{d-d_w(j_{\min})},$$

which is also odd.

Therefore,

$$\nu_2 \left(\delta_{j_{\min}} 2^{j_{\min}-1} \right) = j_{\min} - 1. \quad (5.9)$$

Step 3: Indices $j > j_{\min}$.

For indices

$$j > j_{\min},$$

the coefficient δ_j need not vanish even when

$$v_j = w_j,$$

because the partial Hamming weights $d_v(j)$ and $d_w(j)$ may differ.

No cancellation of these terms is therefore assumed.

However, each term

$$\delta_j 2^{j-1}$$

is divisible by

$$2^{j-1},$$

and since

$$j - 1 \geq j_{\min},$$

we obtain

$$\nu_2 \left(\sum_{j > j_{\min}} \delta_j 2^{j-1} \right) \geq j_{\min}. \quad (5.10)$$

Step 4: Conclusion.

Write

$$Q_k(v^{(k)}) - Q_k(w^{(k)}) = X + Y,$$

where

$$X = \delta_{j_{\min}} 2^{j_{\min}-1}$$

and

$$Y = \sum_{j > j_{\min}} \delta_j 2^{j-1}.$$

By (5.9) and (5.10),

$$\nu_2(X) = j_{\min} - 1 < j_{\min} \leq \nu_2(Y).$$

Therefore, by the non-Archimedean property of the 2-adic valuation,

$$\nu_2(X + Y) = \nu_2(X) = j_{\min} - 1.$$

Since

$$Q_k(v^{(k)}) - Q_k(w^{(k)}) = 2^k \left(q_k(v^{(k)}) - q_k(w^{(k)}) \right),$$

we obtain

$$\nu_2 \left(2^k \left(q_k(v^{(k)}) - q_k(w^{(k)}) \right) \right) = j_{\min} - 1,$$

which proves (5.5).

Finally,

$$2^k \left| q_k(v^{(k)}) - q_k(w^{(k)}) \right| \geq 2^{j_{\min}-1}.$$

Dividing by 2^k gives

$$\left| q_k(v^{(k)}) - q_k(w^{(k)}) \right| \geq 2^{-(k-j_{\min}+1)},$$

which proves (5.6). □

Remark 5.3 (Necessity of the Equal-Weight Hypothesis). The equal-weight assumption in Theorem 5.2 is essential.

For example, let

$$k = 2, \quad v^{(2)} = (1, 0), \quad w^{(2)} = (1, 1).$$

Then

$$j_{\min} = 2,$$

but

$$Q_2(v^{(2)}) = 1$$

and

$$Q_2(w^{(2)}) = 3 + 2 = 5.$$

Therefore,

$$\nu_2(Q_2(v^{(2)}) - Q_2(w^{(2)})) = \nu_2(-4) = 2,$$

whereas

$$j_{\min} - 1 = 1.$$

Thus, the conclusion of Theorem 5.2 does not hold for parity vectors of unequal Hamming weight.

The reason is that when

$$d_v \neq d_w,$$

the terms preceding $j_{\min}$ need not cancel, even though the corresponding prefixes coincide. Their coefficients involve different powers of 3 determined by the distinct total Hamming weights. Consequently, the 2-adic valuation in the unequal-weight case depends on additional arithmetic interactions and is not determined solely by the earliest discrepant index.

A general valuation formula for parity vectors of unequal Hamming weight is not pursued in the present paper.

Corollary 5.4 (Arithmetic Separation within a Fixed (k, d) Layer). *Let*

$$v^{(k)}, w^{(k)} \in H_k$$

be distinct parity vectors of equal Hamming weight d , and let $j_{\min}$ denote the earliest index at which they differ.

Then

$$\nu_2(2^k(q_k(v^{(k)}) - q_k(w^{(k)}))) = j_{\min} - 1.$$

Consequently, within a fixed (k, d) layer, parity vectors sharing a longer common prefix are arithmetically closer in the 2-adic metric, whereas parity vectors differing near the beginning are necessarily more widely separated.

Proof. This follows immediately from Theorem 5.2. □

Remark 5.5. The restriction to a fixed (k, d) layer is natural within the weight-based framework of the present paper. Section 3 organizes parity vectors according to their Hamming weight, while Theorem 5.2 provides an arithmetic comparison for parity vectors having both the same length and the same Hamming weight.

Thus, the theorem reveals an intrinsic arithmetic structure within each fixed (k, d) layer of the parity-vector space.

5.3 Structural Consequences of the 2-adic Rigidity Theorem

Theorem 5.2 shows that, for two distinct parity vectors having the same length and the same Hamming weight, the 2-adic valuation of the difference between their correction terms is completely determined by the earliest position at which the two vectors differ.

Thus, within each fixed (k, d) layer, a direct correspondence exists between a combinatorial property of parity vectors—the length of their common initial prefix—and an arithmetic property of the associated correction terms.

More precisely, if two parity vectors first differ at the index $j_{\min}$, then

$$\nu_2 \left(2^k \left(q_k(v^{(k)}) - q_k(w^{(k)}) \right) \right) = j_{\min} - 1.$$

Consequently, within a fixed (k, d) layer, parity vectors sharing a longer common prefix are arithmetically closer in the 2-adic metric, whereas vectors differing near the beginning are more widely separated in that metric.

This result shows that the correction term is not merely an auxiliary quantity appearing in the fundamental Collatz identity. Within each fixed (k, d) layer, it carries arithmetic information reflecting the prefix structure of the underlying parity vectors.

The arithmetic structure revealed by Theorem 5.2 is complementary to the combinatorial structure established in Theorem 3.2. The two results concern different structural aspects of the parity-vector space.

Theorem 3.2 establishes a combinatorial correspondence between neighboring weight layers through the explicit bijection

$$\Phi : U(d) \longrightarrow J(d+1),$$

whereas Theorem 5.2 establishes an arithmetic rigidity within each fixed (k, d) layer. Thus, the two principal results may be summarized schematically as

$$U(d) \xleftrightarrow{\Phi} J(d+1) \quad (\text{combinatorial correspondence}),$$

and

$$(k, d)\text{-layer} \longrightarrow 2\text{-adic arithmetic rigidity}.$$

Taken together, these results provide complementary combinatorial and arithmetic descriptions of finite parity-vector structures. The geometric interpretation developed in Section 4 provides a common visual framework for their organization on the (k, d) lattice plane.

These finite structural results do not by themselves determine the existence or nonexistence of an infinite non-convergent trajectory. Clarifying whether, and in what manner, the finite combinatorial and arithmetic structures established here are related to the global dynamics of the accelerated Collatz map remains an open problem.

6 Numerical Results

To complement the finite-dimensional combinatorial and arithmetic results established in the preceding sections, large-scale computations were performed in both the conventional length-based framework and the weight-based framework developed in this paper.

The purpose of these computations is to examine the finite-dimensional counting behavior of parity vectors satisfying the prefix condition (3.1). The numerical results reported in this section are observational and are not used in the proofs of the structural identity or the 2-adic rigidity theorem.

6.1 Behavior in the Length Basis

In the conventional classification, parity vectors are grouped according to their length k . Let

$$W(k)$$

denote the number of parity vectors of length k satisfying the prefix condition (3.1), and define the relative density

$$\rho_k = \frac{W(k)}{2^k}. \quad (6.1)$$

Thus, ρ_k represents the proportion of length- k binary parity vectors satisfying the prefix condition.

Representative values are summarized in Table 1.

Table 1: Total number of parity vectors 2^k , number of sequences satisfying the necessary conditions for unconverged growth $W(k)$, and ratio ρ_k in the length k basis

k	2^k	$W(k)$	$\rho_k = W(k)/2^k$
1	2	1	5.00×10^{-1}
10	1024	64	6.25×10^{-2}
100	1.27×10^{30}	3.03×10^{26}	2.39×10^{-4}
1,000	1.07×10^{301}	3.11×10^{282}	2.90×10^{-19}
5,000	1.41×10^{1505}	2.03×10^{1425}	1.44×10^{-80}
10,000	2.00×10^{3010}	4.78×10^{2854}	2.40×10^{-156}

The numerical values show that

$$\rho_k = \frac{W(k)}{2^k}$$

decreases substantially as the sequence length increases over the computed range. In particular,

$$\rho_{10000} = 2.40 \times 10^{-156},$$

illustrating the extreme sparsity, within the full binary space of length 10,000, of parity vectors satisfying the prefix condition.

These computations describe finite-dimensional counting behavior only. No asymptotic conclusion concerning $W(k)$ or ρ_k is inferred from the numerical data alone.

6.2 Behavior in the Weight Basis

The principal framework of the present paper classifies parity vectors according to their Hamming weight d .

Recall that

$$W(d) = |U(d)|$$

denotes the number of unconverged parity vectors of weight d , and let

$$k_{\min}(d) = \lceil \lambda d \rceil$$

denote the minimum integer length at or beyond the critical boundary for weight d . We define the normalized relative density

$$\rho_d = \frac{W(d)}{2^{k_{\min}(d)}}. \quad (6.2)$$

This normalization provides a finite-dimensional scale for examining the growth of $W(d)$ relative to the binary space associated with the critical length scale $k_{\min}(d)$.

Representative values are shown in Table 2.

Table 2: Weight-based statistics: number of unconverged sequences $W(d)$ and normalized density ρ_d

d	$k_{\min}$	$2^{k_{\min}}$	$W(d)$	ρ_d
1	2	4	1	2.50×10^{-1}
2	4	16	2	1.25×10^{-1}
5	8	256	12	4.69×10^{-2}
10	16	65536	961	1.47×10^{-2}
20	32	4.29×10^9	13,472,296	3.14×10^{-3}
100	159	7.31×10^{47}	2.31×10^{42}	3.16×10^{-6}
1,000	1,585	1.36×10^{477}	5.55×10^{448}	4.08×10^{-29}
5,000	7,925	4.60×10^{2385}	5.10×10^{2260}	1.11×10^{-125}
10,000	15,850	2.12×10^{4771}	3.24×10^{4526}	1.53×10^{-245}

The computed values indicate a substantial decrease of the normalized relative density as the Hamming weight increases. In particular,

$$\rho_{10000} = 1.53 \times 10^{-245}.$$

Thus, within the normalization adopted in (6.2), the numerical data exhibit a pronounced decrease over the computed range up to $d = 10,000$.

This behavior is purely numerical at present. In particular, the computations do not establish an asymptotic formula or a rigorous rate of decay for ρ_d .

Figure 3 displays the normalized relative density ρ_d as a function of the Hamming weight d , with ρ_d shown on a logarithmic vertical scale.

Over the computed range, the data exhibit an approximately linear trend on the semi-logarithmic plot. This numerical behavior is consistent with an approximately exponential decrease of ρ_d over the investigated finite range up to $d = 10,000$.

However, this observation should not be interpreted as a proof of asymptotic exponential decay. Determining the asymptotic behavior of $W(d)$ and ρ_d remains an open problem.

6.3 Comparison of the Two Frameworks

The numerical computations reveal decreasing relative densities in both the length-based and weight-based frameworks.

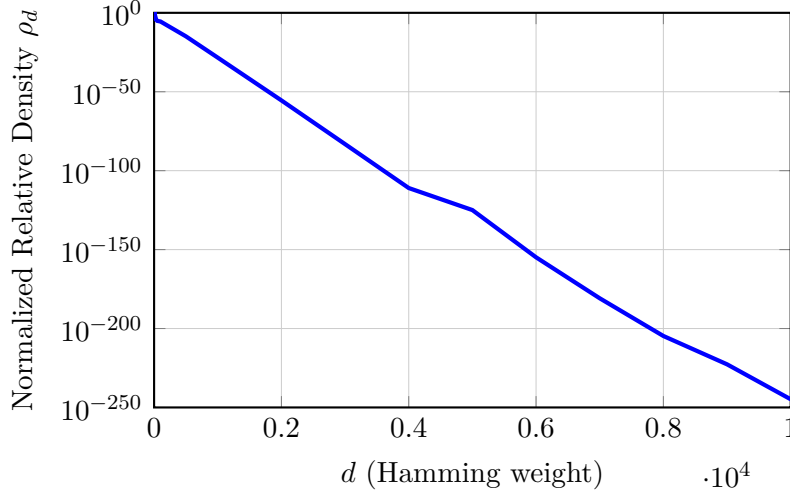


Figure 3: Logarithmic plot of the normalized relative density ρ_d as a function of the Hamming weight d . The vertical axis represents $\log_{10}(\rho_d)$.

In the length-based framework,

$$\rho_k = \frac{W(k)}{2^k}$$

measures the proportion of length- k binary parity vectors satisfying the prefix condition.

In the weight-based framework,

$$\rho_d = \frac{W(d)}{2^{k_{\min}(d)}}$$

provides a normalization of the number of unconverged parity vectors of weight d relative to the binary space associated with the critical length scale $k_{\min}(d)$.

The two quantities are based on different classifications and different normalizing denominators. Therefore, ρ_k and ρ_d should not be interpreted as directly comparable probabilities. Their significance lies instead in describing two distinct finite-dimensional counting frameworks for parity-vector spaces.

The numerical results presented in this section are observational in nature. A rigorous asymptotic analysis of $W(k)$, $W(d)$, ρ_k , and ρ_d remains an open problem.

6.4 Computational Verification and Reproducibility

The numerical computations were primarily carried out using a dynamic programming implementation written in PHP with arbitrary-precision integer arithmetic.

To provide an independent verification of the computational results, the counting procedures were also implemented separately in Python. The two implementations produced identical numerical results over all tested parameter ranges, including the computed values of

$$W(k), \quad W(d), \quad X(d), \quad \rho_k, \quad \text{and} \quad \rho_d.$$

The agreement between the independent implementations provides an additional computational check on the reported numerical results.

The counting algorithms and their pseudocode are described in Appendix A, where the computational procedure and reproducibility of the numerical results are discussed in detail.

7 Discussion and Future Directions

The results established in this paper concern finite-dimensional structures associated with parity vectors of the accelerated Collatz map. In particular, Theorem 3.2 establishes an explicit combinatorial correspondence between unconverged and just-converged parity vectors, whereas Theorem 5.2 reveals an arithmetic rigidity governing differences between correction terms within each fixed (k, d) layer.

These two results provide complementary combinatorial and arithmetic descriptions of finite-dimensional parity-vector spaces. The geometric interpretation developed in Section 4 provides a visual framework for understanding the combinatorial classification on the (k, d) lattice plane. The possible implications of these finite-dimensional structures for the global dynamics of the accelerated Collatz map remain an open problem.

7.1 Relation to the Collatz Conjecture

The present work establishes two principal finite-dimensional structural results.

First, Theorem 3.2 constructs an explicit bijection

$$\Phi : U(d) \longrightarrow J(d+1),$$

thereby proving the structural identity

$$X(d+1) = W(d).$$

This result provides a rigorous combinatorial explanation of the counting correspondence previously observed numerically in the weight-based framework.

Second, Theorem 5.2 gives an exact evaluation of the 2-adic valuation of differences between correction terms associated with parity vectors of the same length and the same Hamming weight. Within each fixed (k, d) layer, the valuation is completely determined by the earliest discrepant index of the corresponding parity vectors, thereby revealing an intrinsic arithmetic rigidity.

These results are finite-dimensional statements derived without assuming either the truth or the falsity of the Collatz conjecture.

The numerical results presented in Section 6 indicate a substantial decrease of the normalized relative density

$$\rho_d = \frac{W(d)}{2^{k_{\min}(d)}}$$

over the computed range up to $d = 10,000$. However, no rigorous connection is currently known between this finite-dimensional numerical behavior and the existence or nonexistence of infinite non-convergent trajectories.

Accordingly, the present work should be regarded as a study of finite-dimensional combinatorial and arithmetic structures of parity-vector spaces rather than as a proof strategy for the Collatz conjecture itself. Whether these structures can be extended or connected to the global dynamics of the accelerated Collatz map remains an open question.

7.2 Open Problems

The results of this paper naturally lead to several open problems.

1. Determine the asymptotic behavior of $W(d)$ and the normalized relative density ρ_d .
2. Determine whether closed formulas, asymptotic formulas, or generating functions for $W(d)$ can be obtained.
3. Investigate further arithmetic consequences of the 2-adic rigidity theorem and the arithmetic structure induced by the correction term.
4. Clarify whether the combinatorial correspondence established by the bijection

$$\Phi : U(d) \longrightarrow J(d+1)$$

admits further structural generalizations.

5. Determine whether the finite-dimensional structures established in this paper can be meaningfully extended to infinite parity vectors and, ultimately, related to the global dynamics of the accelerated Collatz map.

These questions indicate that the finite-dimensional combinatorial and arithmetic structures of parity-vector spaces remain far from being fully understood.

7.3 Future Perspective

The geometric interpretation developed in Section 4 naturally raises the question of how hypothetical infinite non-convergent parity vectors, if they exist, might be organized within the parity-vector space.

Any such parity vector must satisfy the prefix condition

$$j < \lambda d(V|_j)$$

at every finite prefix. Geometrically, its lattice path must therefore remain entirely within the unconverged region, above the critical boundary

$$d = \frac{k}{\lambda},$$

or equivalently to the unconverged side of

$$k = \lambda d.$$

The numerical results of Section 6 indicate that the normalized relative density

$$\rho_d = \frac{W(d)}{2^{k_{\min}(d)}}$$

decreases substantially as the Hamming weight increases over the computed finite range. From a geometric viewpoint, this observation motivates a heuristic picture in which hypothetical non-convergent parity vectors, if they exist, might be confined to an increasingly sparse subset of the unconverged region.

Figure 4 presents this idea schematically as a hypothetical infinite corridor. The figure is intended solely to provide geometric intuition; a precise mathematical characterization of such a structure, or even the existence of such a corridor in a rigorous sense, remains an open problem.

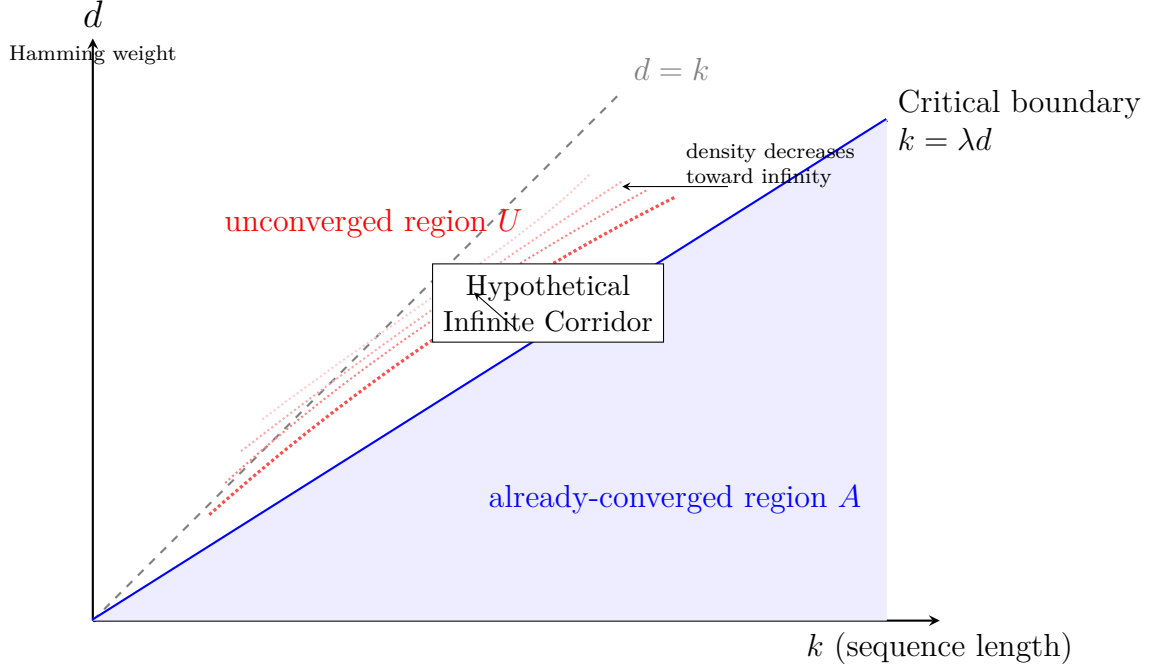


Figure 4: Schematic illustration of a hypothetical infinite corridor inside the unconverged region. If a non-convergent trajectory exists, its parity vector must remain entirely above the critical boundary $k = \lambda d$. The gradual fading of the dotted band is intended to schematically represent the decreasing density suggested by the numerical results. The figure is purely illustrative and should not be interpreted as a rigorous theorem.

Further investigation of the combinatorial and arithmetic organization of the unconverged region may help clarify whether meaningful connections can be established between the finite-dimensional structures developed in this paper and the global dynamics of the accelerated Collatz map.

8 Conclusion

In this paper, we investigated finite-dimensional combinatorial and arithmetic structures associated with parity vectors arising from the accelerated Collatz map, together with their geometric interpretation on the (k, d) lattice plane.

By introducing a classification based on the Hamming weight, we formulated the sets $U(d)$ and $J(d)$ and constructed an explicit mapping

$$\Phi : U(d) \longrightarrow J(d+1).$$

We proved that this mapping is bijective and thereby established the structural identity

$$X(d+1) = W(d)$$

as a finite-dimensional combinatorial theorem.

A geometric interpretation of this correspondence was developed on the (k, d) lattice plane. The parity-vector space was decomposed into the unconverged, just-converged, and already-converged regions, providing a discrete geometric representation of the structural classification and the bijection between neighboring weight layers.

We further investigated arithmetic properties of the correction term appearing in the fundamental Collatz identity. For parity vectors of the same length and the same Hamming weight, an exact formula for the 2-adic valuation of differences between correction terms was obtained. Within each fixed (k, d) layer, the valuation is completely determined by the earliest discrepant index of the corresponding parity vectors. This result reveals an intrinsic arithmetic rigidity within fixed (k, d) layers of the parity-vector space.

Large-scale computations were carried out up to

$$d = 10,000,$$

corresponding to the critical length scale

$$k_{\min}(d) = 15,850.$$

The numerical results exhibit substantial decreases in the relative densities considered in both the length-based and weight-based frameworks. In particular, the weight-based normalized relative density decreases markedly over the computed finite range. A rigorous asymptotic explanation of this numerical behavior remains an open problem.

The results established in this paper are finite-dimensional statements derived independently of any unproven assumptions concerning the Collatz conjecture. No rigorous connection between these results and the existence or nonexistence of infinite non-convergent trajectories is established here. Their implications for the global dynamics of the accelerated Collatz map therefore remain an open problem.

Taken together, the explicit bijection underlying the structural identity and the 2-adic rigidity theorem provide complementary combinatorial and arithmetic descriptions of finite-dimensional parity-vector spaces. The former establishes a correspondence between neighboring weight layers, whereas the latter reveals arithmetic rigidity within each fixed (k, d) layer. The geometric interpretation developed in this paper provides a visual framework for the organization of these structures on the (k, d) lattice plane.

These results provide a unified finite-dimensional framework for the study of parity-vector structures within the weight-based classification. Extending this framework to infinite parity vectors and determining whether it has meaningful implications for the global dynamics of the accelerated Collatz map remain principal challenges for future research.

Acknowledgements

In the preparation of this paper, generative AI tools were used for minor assistance in clarifying the prose and adjusting the structure. All final decisions and responsibilities rest solely with the author.

References

- [1] R. Terras, *A stopping time problem on the positive integers*, Acta Arith. **30** (1976), 241–252.
<https://matwbn.icm.edu.pl/ksiazki/aa/aa30/aa3034.pdf>
- [2] J. C. Lagarias, *The $3x + 1$ problem and its generalizations*, Amer. Math. Monthly **92** (1985), 3–23.
<https://www.cecm.sfu.ca/organics/papers/lagarias/>
- [3] J. Simons and B. de Weger, *Theoretical and computational bounds for m -cycles of the $3x + 1$ problem*, Acta Arith. **117** (2005), 51–70. doi:10.4064/aa117-1-4.
- [4] J. C. Lagarias (Ed.), *The Ultimate Challenge: The $3x+1$ Problem*, American Mathematical Society, Providence, RI, 2010.
<https://bookstore.ams.org/mbk-78>
- [5] T. I. Martiny, *The $3n+1$ Problem: Scope, History and Results*, Ph.D. dissertation, University of Pittsburgh, 2015.
<https://api.semanticscholar.org/CorpusID:118849992>
- [6] J. C. Lagarias, *The $3x+1$ Problem: An Overview*, arXiv preprint, 2021. doi:10.48550/arXiv.2111.02635
- [7] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum Math. Pi, **10**(2022), e12,
doi:10.1017/fmp.2022.8
- [8] D. Bařina, *Improved verification limit for the convergence of the Collatz conjecture*, The Journal of Supercomputing, Vol. 81, 2025, Article 810.
doi:10.1007/s11227-025-07337-0
- [9] E. Roosendaal, *On the $3x + 1$ problem*, Online resource, Available at: <http://www.ericr.nl/wondrous/> (accessed May 2025)
- [10] K. Hikawa and K. Nakanishi, *Parity Vector Analysis in the Study of the Collatz Conjecture*, ResearchGate preprint, January 2026. doi:10.13140/RG.2.2.12065.06240
- [11] K. Nakanishi, *A Structural Study of Parity Vectors in the Collatz Conjecture*, Jxiv preprint, February 2026. doi:10.51094/jxiv.3096

Appendix A Exact Counting Algorithms and Computational Reproducibility

To compute the quantities $W(k)$, $X(k)$, $W(d)$, and $X(d)$ presented in this paper, we employ deterministic dynamic programming algorithms based on arbitrary-precision integer arithmetic. These algorithms exactly enumerate parity vectors satisfying the unconverged condition, namely,

$$j < \lambda d_j$$

for every prefix of length j , where

$$\lambda = \log_2 3.$$

Algorithm 1 presents the basic length-based dynamic programming recurrence and is included primarily to describe the counting mechanism. Algorithm 2 uses the same recurrence and incorporates the post-processing procedure used to obtain the weight-based quantities reported in Section 6.

A.1 Notation and Structural Identity

The quantities used in the algorithms are summarized as follows.

- $W(k)$: number of unconverged parity vectors of length k .
- $X(k)$: number of just-converged parity vectors of length k .
- $W(d)$: number of unconverged parity vectors of Hamming weight d .
- $X(d)$: number of just-converged parity vectors of Hamming weight d .

The structural identity

$$X(d+1) = W(d)$$

is established theoretically in Theorem 3.2 through the explicit bijection

$$\Phi : U(d) \longrightarrow J(d+1).$$

The algorithms below provide deterministic procedures for computing the finite-dimensional counting quantities used in the numerical investigations of Section 6.

A.2 Counting Algorithm 1: Length- k Basis

Let

$$W(k, d)$$

denote the number of unconverged parity vectors of length k and Hamming weight d .

Starting from the initial state, parity vectors satisfying the prefix condition are generated recursively. At each step, appending a symbol 1 increases both the length and the Hamming weight by one, whereas appending a symbol 0 increases only the length. The prefix condition is checked whenever a zero is appended.

Algorithm 1 describes this basic dynamic programming recurrence.

Listing 1: Counting Algorithm 1: Length- k Basis

```

1  // Algorithm 1: Length- $k$  Basis DP Counting
2  // Const  $\lambda := \log 3 / \log 2$ 
3  // Input :  $k_{\max}$ 
4  // Output:  $W[k][d]$ ,  $U[k]$ ,  $X[k]$ 
5
6
7   $W[1][1] \leftarrow 1$ 
8
9  for  $k = 1$  to  $k_{\max}-1$  do
10     for  $d = 1$  to  $k$  do
11         if  $W[k][d] > 0$  then
12              $\text{count} \leftarrow W[k][d]$ 
13
14             // Append 1
15             // (Lemma 3.1 guarantees that appending 1 preserves
16             // persistence)
17              $W[k+1][d+1] \leftarrow W[k+1][d+1] + \text{count}$ 
18
19             // Append 0 and check the unconverged condition
20             if  $(k + 1) < d * \lambda$  then
21                  $W[k+1][d] \leftarrow W[k+1][d] + \text{count}$ 
22             else
23                  $X[k+1] \leftarrow X[k+1] + \text{count}$  // Becomes J-PV
24             end if
25         end if
26     end for
27 end for
28
29 for  $k = 1$  to  $k_{\max}$  do
30      $U[k] \leftarrow \sum_d W[k][d]$ 
31 end for

```

Algorithm 1 is intended primarily to display the underlying recurrence in a transparent form. It directly records the counts $W(k, d)$ and derives the length-based quantities by summation over the Hamming weight.

A.3 Counting Algorithm 2: Length-Based Recurrence with Weight Aggregation

Algorithm 2 uses the same dynamic programming recurrence as Algorithm 1. The distinction lies in its computational purpose: after the length-based counts $W(k, d)$ have been generated, the weight-based quantities are obtained by aggregation over all admissible lengths.

Thus, Algorithm 1 emphasizes the basic recurrence and the length-based counting mechanism, whereas Algorithm 2 explicitly includes the post-processing stage required for the weight-based numerical analysis presented in Section 6.

Since Theorem 3.2 establishes the structural identity

$$X(d+1) = W(d),$$

the weight-based quantities $X(d)$ need not be computed by a separate counting procedure once the values of $W(d)$ have been obtained.

For the complete computation of $W(d)$ for all $d \leq d_{\max}$, the length cutoff $k_{\max}$ must be chosen sufficiently large to cover all admissible lengths for these weight levels. In particular, it is sufficient to take

$$k_{\max} \geq \lceil \lambda d_{\max} \rceil.$$

Listing 2: Counting Algorithm 2: Efficient Length-Based Recurrence (Recommended)

```

1 // Algorithm 2: Efficient Length-Based Recurrence Counting
2 // Const lambda:=log3/log2
3 // Input : k_max, d_max
4 // Output: U[k], X[k], W_total[d]
5 W[1][1] ← 1
6
7 for k = 1 to k_max-1 do
8   for d = 1 to k do
9     if W[k][d] > 0 then
10      count ← W[k][d]
11
12      // Append 1 (Lemma 3.1 guarantees persistence is
      preserved)
13      W[k+1][d+1] ← W[k+1][d+1] + count
14
15      // Append 0
16      if (k + 1) < d * lambda then
17        W[k+1][d] ← W[k+1][d] + count
18      else
19        X[k+1] ← X[k+1] + count // Becomes J-PV
20      end if
21    end if
22  end for
23 end for
24
25 // Post-processing
26 for k = 1 to k_max do
27   U[k] ← sum_d W[k][d]
28 end for
29
30 for d = 1 to d_max do
31   W_total[d] ← sum_k W[k][d]
32 end for

```

The weight-based quantities $W(d)$ are obtained by summing the computed values of $W(k, d)$ over all admissible lengths:

$$W(d) = \sum_k W(k, d).$$

The corresponding values of $X(d)$ are then determined from the structural identity

$$X(d+1) = W(d),$$

which has been established independently of the numerical computations in Theorem 3.2.

A.4 Computational Accuracy and Reproducibility

The numerical results presented in this paper, including

$$\rho_{10,000} \approx 1.53 \times 10^{-245},$$

were obtained using a PHP implementation with the GMP extension for arbitrary-precision integer arithmetic.

The counting procedure is deterministic and all combinatorial counts are computed using exact integer arithmetic. The normalized densities are subsequently obtained from these exact integer counts and the corresponding powers of two.

As an independent computational check, the counting procedures were also implemented separately in Python. The two implementations produced identical values over all tested parameter ranges for the quantities

$$W(k), \quad W(d), \quad X(d), \quad \rho_k, \quad \rho_d.$$

The agreement of the independent implementations provides an additional verification of the reported numerical results. Together with the deterministic recurrence and the pseudocode presented above, this provides a reproducible computational procedure for the numerical investigations reported in Section 6.